

Algebra 1 Practice: Probability

Instructions

Complete in 30 minutes. Show clear work. Give probabilities as simplified fractions unless a percent or decimal is requested.

Reference Facts

- Basic probability:

$$P(A) = \frac{\text{number of outcomes in } A}{\text{number of total outcomes}}.$$

- Complement:

$$P(\text{not } A) = 1 - P(A).$$

- Mutually exclusive events cannot happen at the same time:

$$P(A \text{ or } B) = P(A) + P(B).$$

- General “or” rule:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B).$$

- Independent events:

$$P(A \text{ and } B) = P(A)P(B).$$

- Conditional probability:

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}.$$

Problems

1. A bag contains 6 red marbles, 5 blue marbles, and 4 green marbles. One marble is chosen at random. Find:

$$P(\text{red}), \quad P(\text{not blue}), \quad P(\text{green or blue}).$$

2. A fair number cube is rolled once. Let A be the event of rolling an even number and B be the event of rolling a number greater than 4. Find:

$$P(A), \quad P(B), \quad P(A \text{ and } B), \quad P(A \text{ or } B).$$

3. A spinner has 10 equal sections numbered 1 through 10. Find the probability of spinning:
a multiple of 3, a prime number, a number that is both prime and odd.

4. A coin is flipped and a fair number cube is rolled. Find:

$$P(\text{heads and a 5}), \quad P(\text{tails and an odd number}), \quad P(\text{heads or a 6}).$$

5. Events A and B are independent. If

$$P(A) = \frac{3}{5} \quad \text{and} \quad P(B) = \frac{2}{7},$$

find $P(A \text{ and } B)$.

6. Events C and D are mutually exclusive. If

$$P(C) = \frac{1}{4} \quad \text{and} \quad P(D) = \frac{3}{10},$$

find $P(C \text{ or } D)$ and $P(C \text{ and } D)$.

7. Events M and N have

$$P(M) = 0.42, \quad P(N) = 0.35, \quad P(M \text{ and } N) = 0.12.$$

Find $P(M \text{ or } N)$.

8. A card is chosen from a standard 52-card deck. Find:

$$P(\text{heart}), \quad P(\text{king}), \quad P(\text{heart or king}).$$

9. The table shows survey results from 80 students.

	Plays a Sport	Does Not Play a Sport	Total
Has a Pet	18	22	40
No Pet	12	28	40
Total	30	50	80

Find:

$$P(\text{has a pet}), \quad P(\text{plays a sport}), \quad P(\text{has a pet and plays a sport}).$$

10. Using the table in problem 9, find:

$$P(\text{has a pet} \mid \text{plays a sport}), \quad P(\text{plays a sport} \mid \text{has a pet}).$$

Are these two probabilities equal?

11. Using the table in problem 9, determine whether the events “has a pet” and “plays a sport” are independent. Justify your answer using probabilities.
12. The table shows how 120 students get to school.

	Bus	Car	Walk
Grade 9	28	15	17
Grade 10	22	25	13
Total	50	40	30

Find:

$$P(\text{Grade 9}), \quad P(\text{car}), \quad P(\text{Grade 10 and bus}).$$

13. Using the table in problem 12, find:

$$P(\text{walk} \mid \text{Grade 9}), \quad P(\text{Grade 10} \mid \text{car}).$$

14. A school club has 9 seventh graders and 11 eighth graders. Two students are chosen at random without replacement. Find:

$$P(\text{both are seventh graders}), \quad P(\text{one seventh grader and one eighth grader}).$$